



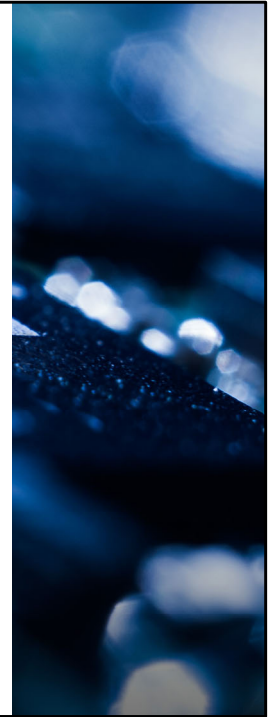
# SC1006

## Computer Organisation and Architecture

### Volatile Memory

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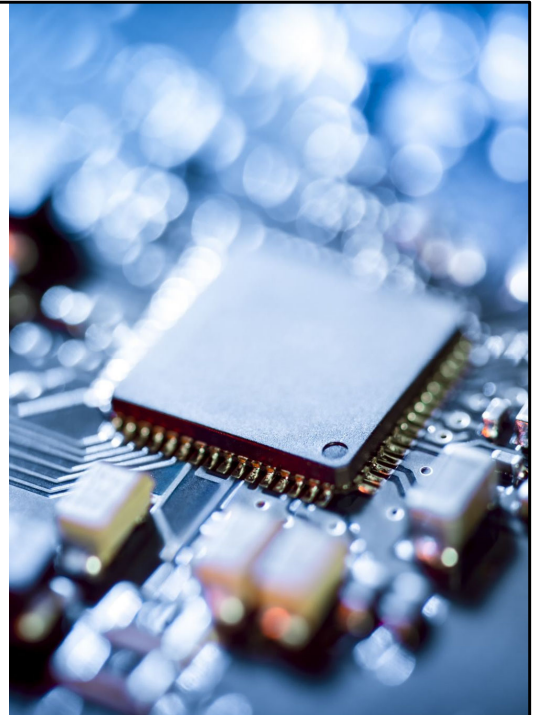
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- This section is an introduction on **Volatile Memory**

## Random Access Memory (RAM)

- Static RAM (SRAM)
  - Static Random Access Memory
  - Data stored as long as supply is applied.
  - Large (4 to 6 transistors per cell).
  - Fast (SRAM is optimized for speed/access-latency)
  - Low power consumption (active and standby)
- Dynamic Random Access Memory (DRAM)
  - Periodic refresh required.
  - Small (1 to 3 transistors per cell).
  - Slower (DRAM is optimized for density/higher-capacity; refresh takes time)
  - Higher power consumption due to need for periodic refresh operation to maintain data integrity of memory cells.



In this course, we will look at two types of volatile memory, SRAM and DRAM.

Static Random Access Memory, SRAM in short

Is able to retain its data integrity as long as power supply is maintained.

It has a relatively larger layout consisting of 4-6 transistors per cell, typically.

Its access time is fast

And power consumption for both active and standby mode is low.

Dynamic RAM, DRAM in short is another type of volatile memory you will frequently encounter.

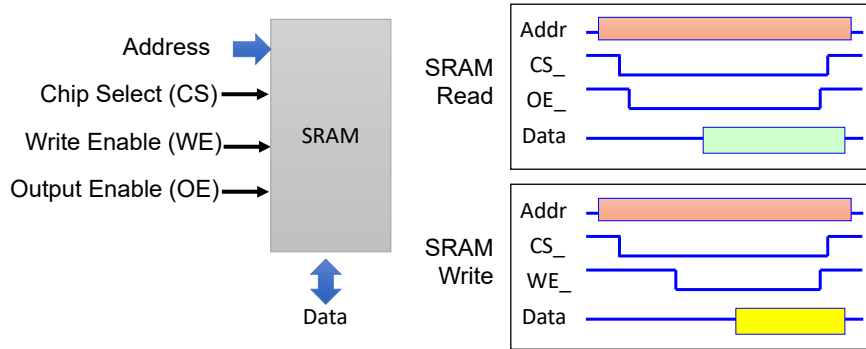
It consists of a capacitor and one transistor.

This gives rise to a smaller layout compared to SRAM, and cost per bit of DRAM is therefore lower than SRAM

The use of capacitor as a storage element, however, requires the cell to be refreshed periodically in order to maintain data integrity.

That leads to a higher power consumption and slower access time when compared to SRAM.

## Static RAM (SRAM) Access



|             |                                       |                                             |
|-------------|---------------------------------------|---------------------------------------------|
| <b>Addr</b> | Specifies Address of memory location. |                                             |
| <b>Data</b> | Data to be read/written.              | Typically 8,16 or 32 bits.                  |
| <b>CS</b>   | Control signals                       | Chip Select (Enables or disables the chip). |
| <b>WE</b>   |                                       | Write Enable (Allows data to to be written) |
| <b>OE</b>   |                                       | Output Enable (Allows SRAM to output data)  |

To perform read and write operation on the SRAM, we need the following signal lines.

Address, which specify which memory location will be read or written

Chip select which is the main switch for the SRAM chip

The Write Enable pin which tells the SRAM that the processor is performing a write operation

The Output Enable pin which allows the SRAM to output data from its data bus.

And lastly the data bus.

During a SRAM read, the uP will first output the address info, followed by the CS and OE. The SRAM will then output the data requested on the data bus.

I'm showing these signaling sequentially to walk you thru the whole process. In actual case, these signaling happen simultaneously.

Similarly for SRAM write, uP wil first output the Address, followed by the CS and WE so that SRAM can prepare for a write operation. uP will then output the data on the data bus and SRAM will latch the information in.

And again, these signals are output simultaneously.

## SRAM Chip DeCap

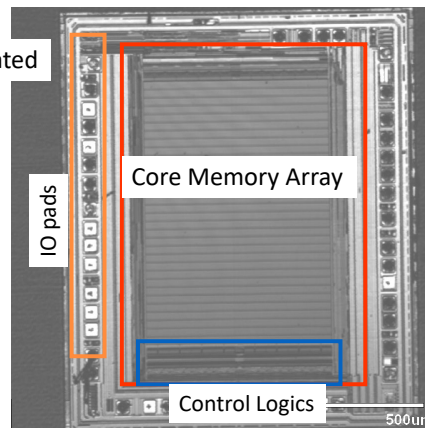
Commercial SRAM Chip  
Cypress CY62128, 1 Mbit



Decapsulated



Decapsulated silicon die  
Laser scanned image, 5x magnification

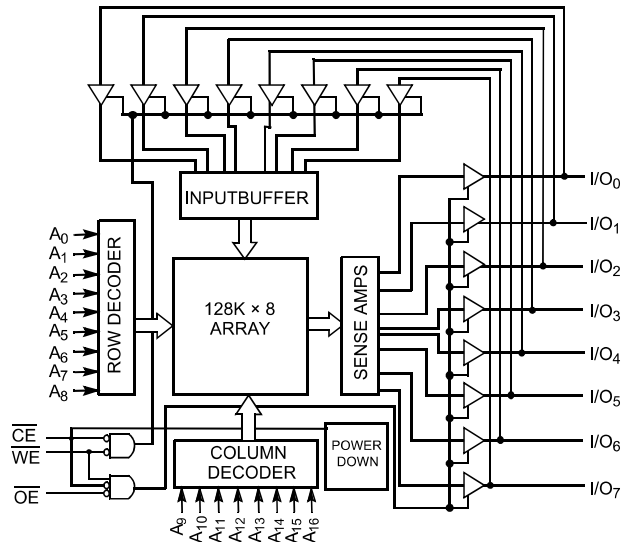


- **Memory** array takes up **most** of the **real estate**.
- Leading Edge Semiconductor Process is usually first tested with Memory Design.

- Next, let's dig deeper into the SRAM chip.
- This show what it looks like under the packaging. You can see rows and rows of uniform design, which is actually the SRAM cells.
- In most system or uP, memory usually takes up the most real estate.
- As such, they are usually the target candidate to test the leading edge semiconductor process technology.
- Which is why leading memory makers such as Samsung are usually the leaders in semiconductor process technology.
- The other group of companies that leads in the process technology are the processor makers such as Intel and AMD.

## SRAM Internal Circuitry

- Memory array
- 1M SRAM cells for the chip shown.
- Control circuitry
- Decoders
- Sense amplifiers
- Input/Output multiplexers



This is how SRAM internal looks like functionally.  
The example shown here is a 128KByte SRAM chip

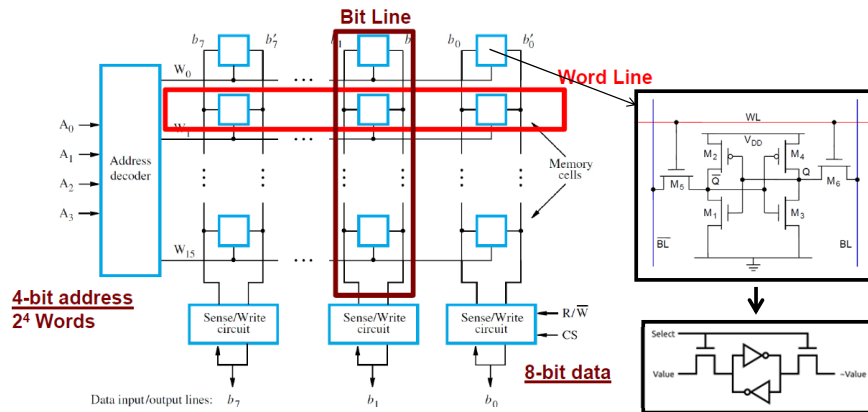
It consists of an array of 1 Mega ( $2^{20}$ ) SRAM memory cells, and each cell contains 1 bit.

The addresses are fed into decoders which will enable 8 of the cells (for a byte) so that read or write operation can be carried out.

For SRAM, you don't have to be too concerned about the detail design such as the role of decoders, multiplexers and sense amplifiers; they are not in the syllabus. We explain more of SRAM in the next slides.

# SRAM Cell

- Memory cells are organized in arrays (rows), and are accessed via the **word lines** and **bit lines**.
- Each individual SRAM Bit consist of 6 Transistors (typical design).



Diving in further, this is how SRAM cells are arranged.

The address info is decoded to enable a particular word of data, each word consists of one row of cells, each cell corresponding to one bit of data.

Each SRAM cell has six transistors. It has a differential output.

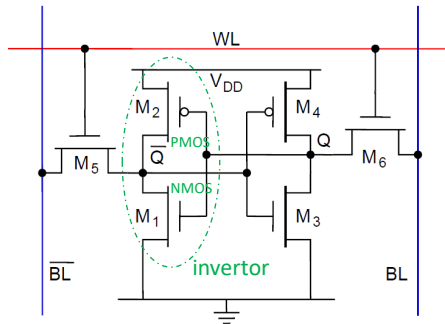
The digital logic equivalence of the transistor design is shown here.

It is basically two inverters with output of one connected to the input of the other.

This ensures that the logic state that is stored will remain unchanged as long as the gates are powered.

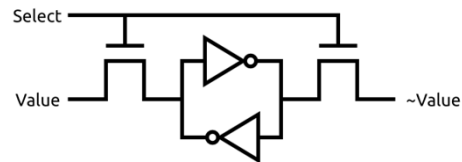
I do not expect you to know the detail transistor level layout design but you should at least know the function of the transistors and the logic equivalent of the 6-transistor design, as to be explained more in the next slides.

# SRAM Cell



- There are six transistors
- M1, M2, M3, M4, M5 and M6
- Word line (WL) is derived from Address Decoder Output. It selects the word location for read and write
- The actual data are placed on the differential bit lines (BL and BLB). This is connected to the Data Bus.

- The data bit is stored in M1, M2, M3 and M4 (which is equivalent to two inverters connected as shown on the right).
- M5 and M6 are pass (switch) transistors.



As mentioned in the previous slide, a SRAM cell consist of 6 transistors.

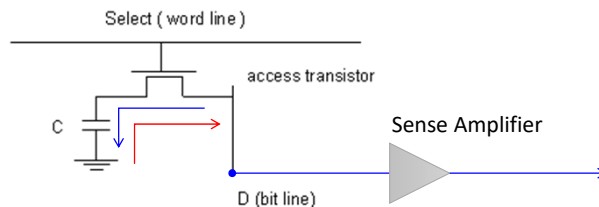
The word line is derived from the address info and controls which sets of bit cells are enabled in the word.

The bit line output the actual data.

M1 to M4 forms the inverter logic we mentioned earlier.

M5 and M6 controls the cell to be connected for read/write operation.

# Dynamic RAM (DRAM)



- Single Transistor Design
- DRAM uses a capacitor as its storage element.
- The Transistor is used to control charges flowing in and out of the capacitor during the Read and Write process.
- Write Process
  - To store a Logic '1': Enable the Transistor, transfer charge into capacitor.
  - To store a Logic '0': Enable the Transistor, discharge the capacitor.
- Read Process
  - Enable the Transistor. Measure the capacitor charge using a sense amplifier.

Next we are going to look at Dynamic RAM. DRAM in short.

DRAM cell only consists of one single transistor and a capacitor.

The capacitor is used to store the charge while the transistor controls the flow of charges in and out of the capacitor.

First, let's look at the Write Process

To store a logic '1', we first enable the pass transistor and transfer some charge to the capacitor to charge it up.

If we want to write a logic '0' instead, then we would need to enable the pass transistor and discharge the capacitor.

Similarly, for the Read Process

The pass transistor must be enabled and the capacitor charge status is detected using a sense amplifier, in order to determine the logic stored in the DRAM cell.

As can be seen, DRAM is much simpler and therefore cheaper than SRAM.



## DRAM – Maintaining Data Integrity

- DRAM Read Process destroys information stored on capacitor
- The process of measuring charges on a capacitor also effectively discharged it i.e. data is destroyed.
- Hence, the original data has to be re-written back after every read.
- Periodic refresh is needed as the stored charge “leaks” with time.
- The basic DRAM is more or less obsolete in the market today. It is replaced by its synchronous version called Synchronous DRAM (SDRAM).
- Difference between SDRAM and DRAM is that the former makes use of a clock signal from the host to synchronize data transfer, enabling faster transfer rate. SDRAM also has a pipeline architecture that allow faster, overlapping operations.
- Other enhancements of SDRAM includes its double data rate versions DDR, DDR2, DDR3 SDRAM, which could reach transfer speed of more than 2G transfers per second.

DRAM is called Dynamic RAM because the charges stored in the DRAM cell tends to change; this is compared to the SRAM, which can hold its charges indefinitely as long as there is power supplied.

Let’s look at this behaviour in more detail.

In DRAM, a read process involves discharging the capacitor so in a way, the data stored is destroyed when performing the read operation.

That’s why the original data must be re-written after every read.

On top of that, charges in a capacitor tend to leak with time. Hence, DRAM cells requires periodic refresh operation to hold the data.

Note that the basic DRAM is less-used in the market today. It is replaced by the synchronous version called Synchronous DRAM, or SDRAM in short.

But basic concepts and features of the DRAM remains, such as the use of capacitor as main storage element and the need to have periodic refresh to maintain data

integrity.

The difference between SDRAM and DRAM is that

SDRAM uses a clock signal from host to synchronise the data transfer  
SDRAM also employs a pipeline architecture that allows faster, overlapping operations compared to conventional DRAM.

The SDRAM you see today is enhancement over the single rate SDRAM. Known as DDR2, DDR3, DDR4 etc which can sustain higher data transfer rate.

That conclude the discussion on volatile memory.

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